1 Mdm Wang runs a stall in a food court selling three types of noodles: wonton noodles, fishball noodles and beef noodles. For each bowl of wonton noodles, fishball noodles and beef noodles, she makes a profit of exactly 60 cents, 90 cents and \$1.20 respectively.

On a particular day, Mdm Wang made a total profit of \$222.00, and she earned \$72.00 more in profit from the sale of beef noodles than from wonton noodles. The total number of bowls of noodles sold that day was 4 times the number of bowls of fishball noodles sold that day.

(a) Find the number of bowls of wonton noodles, fishball noodles and beef noodles that Mdm Wang sold that day. [3]

Due to a system error, all customers who made online payments for wonton and beef noodles on that day were overcharged. Customers were charged \$5.50 instead of the correct price of \$4.50 for a bowl of wonton noodles, and \$7.00 instead of \$6.50 for a bowl of beef noodles. It was found that 1 in 4 bowls of wonton noodles and 1 in 5 bowls of beef noodles were paid for online.

(b) Find the total amount Mdm Wang had to refund to all affected customers. [1]

Q1	**Solution**
(a)	Let w, f and b be the number of bowls of wonton noodles, fishball noodles and beef noodles sold respectively. $0.6w + 0.9f + 1.2b = 222$ ... ① $1.2b - 0.6w = 72$ ... ② $w + f + b = 4f \Rightarrow w - 3f + b = 0$ ... ③ Using GC, $w = 80, f = 60, b = 100$ Mdm Wang sold 80 bowls of wonton noodles, 60 bowls of fishball noodles and 100 bowls of beef noodles.
(b)	Total refund $= \left(\frac{1}{4}\times 80\right)\times \$1 + \left(\frac{1}{5}\times 100\right)\times \$0.50 = \$30$

2 A curve C has equation $xy = \ln y$, where $y > 0$.

(a) Find $\frac{dy}{dx}$ in terms of x and y, and hence show that C has no stationary points. [3]

(b) Find the equation of the tangent to C at the point where $y = e^2$. Give your answer in the form $y = mx + c$, where m and c are exact values. [3]

Q2	[Solution]
(a)	$xy = \ln y$ Differentiating w.r.t. x, $(1)\ y + x\frac{dy}{dx} = \frac{1}{y}\frac{dy}{dx}$ $\Rightarrow \quad \left(\frac{1}{y} - x\right)\frac{dy}{dx} = y$ $\Rightarrow \quad \frac{dy}{dx} = \frac{y^2}{1 - xy}$ Since $y > 0$, $\frac{dy}{dx} \neq 0$ **at all points** on curve. Thus C has no stationary points. (Shown)
(b)	When $y = e^2$, $x = \frac{\ln e^2}{e^2} = \frac{2}{e^2}$ $\frac{dy}{dx} = \frac{\left(e^2\right)^2}{1 - \left(\frac{2}{e^2}\right)\left(e^2\right)} = -e^4$ Equation of tangent to C at point where $y = e^2$ is $y - e^2 = -e^4\left(x - \frac{2}{e^2}\right)$ $y = -e^4 x + 3e^2$

3 Solve the inequality $\frac{4}{x-\lambda} < x-\lambda$, where λ is a real constant. Give your answer in terms of λ.

[4]

Q3	Solution

$$\frac{4}{x-\lambda} < x-\lambda$$

$$\frac{4-(x-\lambda)^2}{x-\lambda} < 0$$

$$\frac{(x-\lambda)^2-4}{x-\lambda} > 0$$

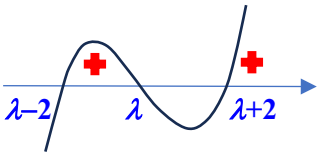
Multiply by $(x-\lambda)^2$ on both sides,

$$(x-\lambda)\left[(x-\lambda)^2-4\right] > 0$$

Let $(x-\lambda)^2-4=0$ (to find the x-intercepts)

$$(x-\lambda)^2=4 \Rightarrow x-\lambda=\pm 2 \Rightarrow x=\lambda\pm 2$$

For $(x-\lambda)(x-(\lambda-2))(x-(\lambda+2))>0$



From the graph, $\lambda-2<x<\lambda$ or $x>\lambda+2$

Alternative Solution

$$\frac{4}{x-\lambda} < x-\lambda$$

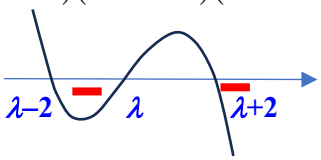
Multiply by $(x-\lambda)^2$ on both sides,

$$4(x-\lambda) < (x-\lambda)^3$$

$$4(x-\lambda)-(x-\lambda)^3 < 0$$

$$(x-\lambda)\left(4-(x-\lambda)^2\right) < 0$$

$$(x-\lambda)(2+x-\lambda)(2-x+\lambda) < 0$$



From the graph, $\lambda-2<x<\lambda$ or $x>\lambda+2$

Not so efficient method

$$\frac{4}{x-\lambda} < x-\lambda$$

$$\frac{4-(x-\lambda)^2}{x-\lambda} < 0$$

$$\frac{4-\left(x^2-2\lambda x+\lambda^2\right)}{x-\lambda} < 0$$

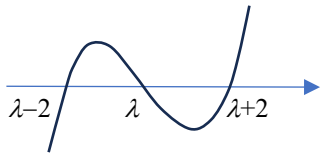
$$\frac{-x^2+2\lambda x+4-\lambda^2}{x-\lambda} < 0$$

$$\frac{x^2-2\lambda x+\lambda^2-4}{x-\lambda} > 0$$

Let $x^2-2\lambda x+\lambda^2-4=0$

$$x=\frac{2\lambda\pm\sqrt{(2\lambda)^2-4(1)\left(\lambda^2-4\right)}}{2}=\frac{2\lambda\pm\sqrt{16}}{2}=\lambda\pm 2$$

For $(x-\lambda)(x-(\lambda-2))(x-(\lambda+2))>0$



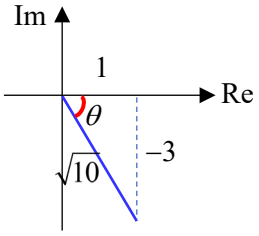
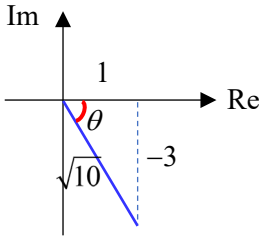
From the graph, $\lambda-2<x<\lambda$ or $x>\lambda+2$

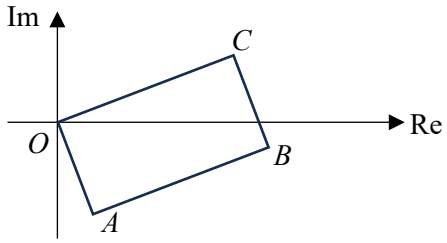
4 **Do not use a calculator in answering this question.**

The complex number z is such that $|z| = \sqrt{10}$ and $\arg(z) = -\tan^{-1} 3$.

(a) Find z in the form $a + bi$ where a and b are integers. [2]

(b) In an Argand diagram, O is the origin and the point A represents z. The points B and C are such that $OABC$ is a rectangle described in an anti-clockwise sense and $OC = 2OA$. Find the complex numbers represented by B and C. [4]

Q4	Solution
(a)	$\arg z = \theta = -\tan^{-1} 3$ (point rep z lies in 4th quadrant) **Method 1 (most efficient method)** $\tan\theta = \frac{-3}{1}$ $\|z\| = r = \sqrt{1^2 + (-3)^2} = \sqrt{10}$  From diagram, $z = 1 - 3i$ **Method 2 (most common method)** Let $z = a + bi$ where ***a* > 0 and *b* < 0** $\|z\| = \sqrt{a^2 + b^2} = \sqrt{10} \Rightarrow a^2 + b^2 = 10$ --- (1) $\tan\theta = \frac{b}{a} = -3 \Rightarrow b = -3a$ --- (2) Subst (2) into (1): $a^2 + 9a^2 = 10$ $\Rightarrow a = 1 \quad (\because a > 0)$ $b = -3$ Thus $z = 1 - 3i$ **Method 3** From diagram, $a = \sqrt{10}\cos\theta = \sqrt{10}\left(\frac{1}{\sqrt{10}}\right) = 1$ $b = \sqrt{10}\sin\theta = \sqrt{10}\left(-\frac{3}{\sqrt{10}}\right) = -3$  Thus $z = 1 - 3i$

(b)	Let complex numbers represented by B and C be b and c respectively.  Since $OC = 2OA$ and OA is perpendicular to OC, $c = 2\,\mathrm{i}z$ $= 2\mathrm{i}(1-3\mathrm{i})$ $= 6+2\mathrm{i}$ By vector addition, $b = z + c$ $= (1 - 3\mathrm{i}) + (6 + 2\mathrm{i})$ $= 7 - \mathrm{i}$

5 It is given that $y = \mathrm{f}(x)$ is such that $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \dfrac{\mathrm{d}y}{\mathrm{d}x}\left(2\tan x + p\sec^2 x\right)$, where p is a constant. The Maclaurin series for $\mathrm{f}(x)$ is given by $y = 1 + x + \dfrac{1}{2}x^2 + qx^3 + \ldots$, where q is a constant.

(a) State the values of $\mathrm{f}(0)$ and $\mathrm{f}'(0)$. [1]

(b) Find the values of p and q. [5]

(c) Using the given series for $\mathrm{f}(x)$ and standard series from the List of Formulae (MF27), find the first three non-zero terms of the series expansion of $\sqrt{\mathrm{f}(x)}$. [2]

(d) The given Maclaurin series for $\mathrm{f}(x)$ is used to estimate $\int_0^{0.2} \mathrm{f}(x)\,\mathrm{d}x$ and $\int_{0.2}^{0.4} \mathrm{f}(x)\,\mathrm{d}x$. Without any further computations, determine which estimate has a smaller percentage error. Give an explanation for your choice. [1]

Q5	[Solution]
(a)	$\mathrm{f}(0) = 1$ and $\mathrm{f}'(0) = 1$
(b)	When $x = 0$, $\dfrac{\mathrm{f}''(0)}{2!} = \dfrac{1}{2}$ and $\dfrac{\mathrm{f}'''(0)}{3!} = q$ $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \dfrac{\mathrm{d}y}{\mathrm{d}x}\left(2\tan x + p\sec^2 x\right)$ When $x = 0$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = (1)\left(0 + p\sec^2 0\right) = \dfrac{p}{\cos^2 0} = p$ Thus $\dfrac{p}{2!} = \dfrac{1}{2} \Rightarrow p = 1$ Differentiating w.r.t. x, $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3} = \dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(2\tan x + p\sec^2 x\right)$ $+ \dfrac{\mathrm{d}y}{\mathrm{d}x}\left(2\sec^2 x + 2p\sec x\left(\sec x\tan x\right)\right)$ When $x = 0$, $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3} = (1)\left(0 + p\sec^2 0\right) + (1)\left(2\sec^2 0 + 0\right) = 1 + 2 = 3$ Thus $q = \dfrac{3}{3!} = \dfrac{1}{2}$
(c)	$\sqrt{\mathrm{f}(x)} = \left(1 + x + \dfrac{1}{2}x^2 + \dfrac{1}{2}x^3 + \ldots\right)^{\frac{1}{2}}$ $= 1 + \dfrac{1}{2}\left(x + \dfrac{1}{2}x^2 + \dfrac{1}{2}x^3 + \ldots\right)$ $+ \dfrac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}\left(x + \dfrac{1}{2}x^2 + \dfrac{1}{2}x^3 + \ldots\right)^2 + \ldots$

	$= 1 + \frac{1}{2}x + \frac{1}{4}x^2 - \frac{1}{8}x^2 + ...$ $= 1 + \frac{1}{2}x + \frac{1}{8}x^2 + ...$
(d)	The estimate for $\int_0^{0.2} \mathrm{f}(x)\,\mathrm{d}x$ will give a smaller percentage error because the Maclaurin series curve is closer to the curve $y = \mathrm{f}(x)$ for x-values closer to zero.

6 The function f is defined by

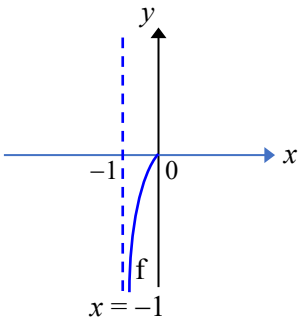
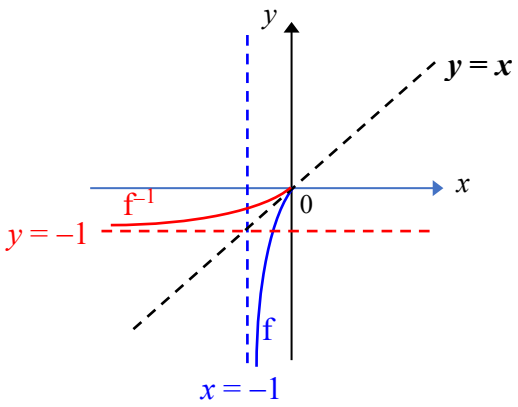
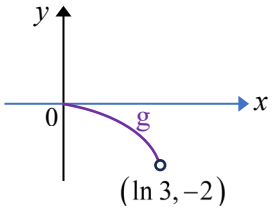
$$\mathrm{f}: x \mapsto \frac{2x}{x+1},\ x \in \mathbb{R},\ -1 < x \leq 0.$$

(a) Find $\mathrm{f}^{-1}(x)$ and state its domain. [3]

(b) Sketch, on the same diagram, the graphs of f and f^{-1}, showing clearly the relationship between the two graphs and giving the equations of the asymptotes. [3]

The function g is defined by $\mathrm{g}: x \mapsto 1 - \mathrm{e}^{x},\ x \in \mathbb{R},\ 0 \leq x < \ln 3$.

(c) Explain why $\mathrm{f}^{-1}\mathrm{g}$ exists. [2]

Q6	Solution
(a)	Let $y = \frac{2x}{x+1}, x \in \mathbb{R}, -1 < x \leq 0$ $(x+1)y = 2x$ $(y-2)x = -y$ $x = \frac{y}{2-y}$ $\therefore\ \mathrm{f}^{-1}(x) = \frac{x}{2-x},\ x \in \mathbb{R},\ x \leq 0$ Domain of f^{-1} = Range of f = $(-\infty, 0]$ 
(b)	
(c)	Given $\mathrm{g}: x \to 1 - \mathrm{e}^{x}\ \ x \in \mathbb{R}, 0 \leq x < \ln 3$ $\mathrm{g}(\ln 3) = 1 - \mathrm{e}^{\ln 3} = 1 - 3 = -2$ Since $\mathrm{R_g} = (-2, 0] \subseteq (-\infty, 0] = \mathrm{D_{f^{-1}}}$, $\mathrm{f}^{-1}\mathrm{g}$ exists. 

7 **(a)** The sum of the first n terms of a series is $\frac{1}{4}\left(13n-n^2\right)$, where n is a positive integer.

(i) Find an expression for u_n, the nth term of the series. [2]

(ii) Show that this series is an arithmetic series. [2]

(b) A convergent geometric series has first term 3 and fourth term $24x^{-3}$, where $x \in \mathbb{R},\ x \neq 0$.

(i) Find the range of values that x can take. [3]

(ii) Find, in terms of x, the sum of the geometric series after but not including the 10th term. Give your answer in the form $\frac{p}{x^m(x-q)}$, where m, p and q are positive integers. [3]

Q7	[Solution]
(a) (i)	Given: $S_n = \frac{1}{4}\left(13n-n^2\right)$ $u_n = S_n - S_{n-1}$ $= \frac{1}{4}\left(13n-n^2\right) - \frac{1}{4}\left(13(n-1)-(n-1)^2\right)$ $= \frac{13}{4}\left(n-(n-1)\right) - \frac{1}{4}\left(n^2-(n-1)^2\right)$ $= \frac{13}{4} - \frac{1}{4}(2n-1)$ $= \frac{14-2n}{4} = \frac{7-n}{2}$
(a) (ii)	$u_n - u_{n-1} = \frac{7-n}{2} - \frac{7-(n-1)}{2}$ $= -\frac{1}{2}$, a constant independent of n Thus the series is an arithmetic series. (Shown)
(b) (i)	$T_4 = 3r^3 = 24x^{-3}$, where r is the common ratio $\Rightarrow \quad r^3 = 8x^{-3}$ $\Rightarrow \quad r = 2x^{-1}$ Since the geometric series is convergent, $\lvert r\rvert < 1$ i.e., $\left\lvert\frac{2}{x}\right\rvert < 1$ $\lvert x\rvert > 2$ $\therefore\ x < -2$ or $x > 2$

(b) **(ii)**	Required sum $= T_{11} + T_{12} + ...$ $= \frac{T_{11}}{1-r} = \frac{3\left(2x^{-1}\right)^{10}}{1-2x^{-1}}$ $= \frac{3072x^{-10}}{1-2x^{-1}}$ $= \frac{3072}{x^{10}\left(\frac{x-2}{x}\right)}$ $= \frac{3072}{x^{9}\left(x-2\right)}$ **Alternative solution** Required sum $= S_{\infty} - S_{10}$ $= \frac{3}{1-r} - \frac{3\left(1-r^{10}\right)}{1-r}$ $= \frac{3r^{10}}{1-r} = \frac{3\left(2x^{-1}\right)^{10}}{1-2x^{-1}}$ $= \frac{3072x^{-10}}{1-2x^{-1}}$ $= \frac{3072}{x^{10}\left(\frac{x-2}{x}\right)}$ $= \frac{3072}{x^{9}\left(x-2\right)}$

8 A plane π_1 is parallel to the two vectors $\begin{pmatrix}2\\1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\\2\end{pmatrix}$, and passes through the point A with coordinates (1, 0, 1).

(a) Show that the cartesian equation of π_1 is $x - 2y + z = 2$. [3]

(b) The line l contains the point B with coordinates (2, 1, 0) and is parallel to the vector $\begin{pmatrix}1\\2\\k\end{pmatrix}$.

Given that the acute angle between l and π_1 is θ where $\sin\theta = \frac{1}{3}$, find the possible values of k. [4]

(c) Given now that $k = 1$, find the coordinates of the point of intersection between l and π_1. [3]

(d) A plane π_2 contains the point C with coordinates (2, 1, –3) and is parallel to π_1. Find the exact distance between π_1 and π_2. [3]

Q8	Solution
(a)	$\begin{pmatrix}2\\1\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\2\end{pmatrix} = 2\begin{pmatrix}1\\-2\\1\end{pmatrix}$ A vector normal to π_1 is $\begin{pmatrix}1\\-2\\1\end{pmatrix}$ $\pi_1 : \underset{\sim}{r}\cdot\begin{pmatrix}1\\-2\\1\end{pmatrix} = \begin{pmatrix}1\\0\\1\end{pmatrix}\cdot\begin{pmatrix}1\\-2\\1\end{pmatrix} = 2$ Cartesian equation of π_1 is $x - 2y + z = 2$
(b)	$\sin\theta = \dfrac{\left\lvert\begin{pmatrix}1\\2\\k\end{pmatrix}\cdot\begin{pmatrix}1\\-2\\1\end{pmatrix}\right\rvert}{\left\lvert\begin{pmatrix}1\\2\\k\end{pmatrix}\right\rvert\left\lvert\begin{pmatrix}1\\-2\\1\end{pmatrix}\right\rvert}$ $\frac{1}{3} = \dfrac{\lvert 1-4+k\rvert}{\left\lvert\sqrt{1^2+(2)^2+k^2}\right\rvert\left\lvert\sqrt{1^2+(-2)^2+1^2}\right\rvert}$ $\frac{1}{3} = \dfrac{\lvert k-3\rvert}{\left\lvert\sqrt{5+k^2}\right\rvert\left\lvert\sqrt{6}\right\rvert}$ Squaring both sides, $9(k-3)^2 = 6(5+k^2)$ $k^2 - 18k + 17 = 0$ $(k-1)(k-17) = 0$ $\Rightarrow k = 1$ or 17

Q8	Solution
(c)	$\pi_1 : x - 2y + z = 2$ i.e. $\underset{\sim}{r} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 2 \ \ldots ①$ $l : \underset{\sim}{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R} \ \ldots ②$ Substitute ② into ①, we have $\begin{pmatrix} 2+\lambda \\ 1+2\lambda \\ \lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 2$ $2 + \lambda - 2 - 4\lambda + \lambda = 2$ $\lambda = -1$ Hence the coordinates of the point of intersection is $(1, -1, -1)$
(d)	The distance from origin to the plane π_1 is $\frac{2}{\sqrt{6}}$ $\pi_2 : \underset{\sim}{r} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = -3$ The distance from origin to the plane π_2 is $\frac{\lvert -3 \rvert}{\sqrt{6}} = \frac{3}{\sqrt{6}}$ Hence the distance between π_1 and π_2 $= \frac{3}{\sqrt{6}} + \frac{2}{\sqrt{6}}$ $= \frac{5}{\sqrt{6}}$ or $\frac{5\sqrt{6}}{6}$

9 The curves C_1 and C_2 have equations

$$y = \frac{ax}{x+2} \text{ and } y = \frac{x^2 + (a+4)x + 2a + 8}{x+2}$$

respectively, where a is a constant.

(a) Find the set of values of a such that C_1 and C_2 do not intersect for all real values of x. [3]

It is now given that $0 < a < 4$.

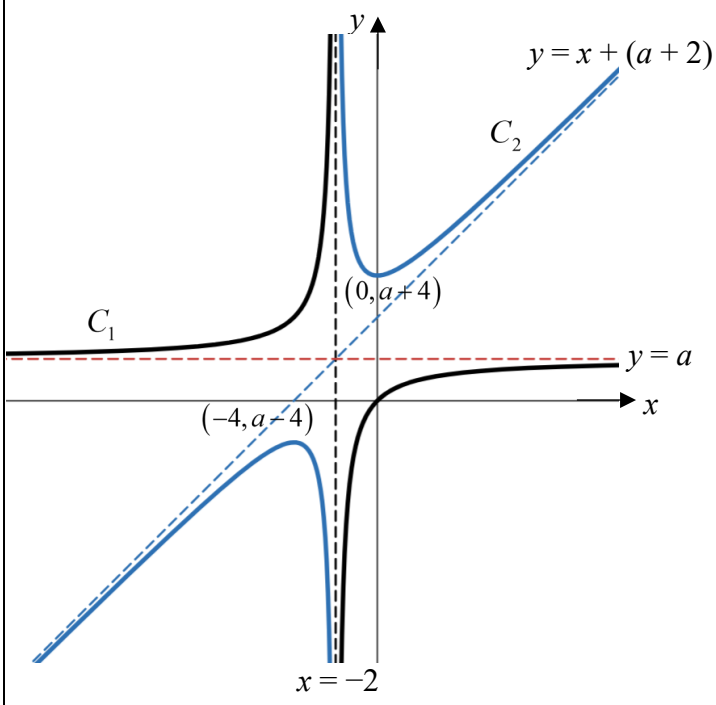
(b) Find, in terms of a, the coordinates of the stationary points of C_2. [4]

(c) Sketch C_1 and C_2 on a single diagram and give the equations of the asymptotes, in terms of a where appropriate. [5]

(d) Hence solve the inequality $\frac{ax}{x+2} > \frac{x^2 + (a+4)x + 2a + 8}{x+2}$. [1]

Q9	**[Solutions]**
(a)	Let $\frac{ax}{x+2} = \frac{x^2 + (a+4)x + 2a + 8}{x+2}$ $ax = x^2 + (a+4)x + 2a + 8$ $x^2 + 4x + 2a + 8 = 0$ ------- (*) Since C_1 and C_2 do not intersect, equation (*) has no real roots. Thus discriminant < 0 $4^2 - 4(1)(2a+8) < 0$ $-8a - 16 < 0$ $a > -2$ Set of values = $\{a : a \in \mathbb{R}, a > -2\}$ Note that when $a = -2$, the 2 curves also do not intersect.
(b)	$\frac{dy}{dx} = \frac{(x+2)(2x+a+4) - [x^2 + (a+4)x + 2a + 8]}{(x+2)^2}$ OR: differentiate $y = x + a + 2 + \frac{4}{x+2}$ Let $\frac{dy}{dx} = 0$ $(x+2)(2x+a+4) - [x^2 + (a+4)x + 2a + 8] = 0$ $2x^2 + (a+4)x + 4x + 2(a+4) - x^2 - (a+4)x - 2(a+4) = 0$ $x^2 + 4x = 0$ $x(x+4) = 0$ $x = 0$ or $x = -4$ When $x = 0$, $y = a + 4$ When $x = -4$, $y = a - 4$ Coordinates of stationary points = $(0, a+4)$ and $(-4, a-4)$

(c)



For C_1: $y = \dfrac{ax}{x+2} = a - \dfrac{2a}{x+2}$

Equations of asymptotes: $y = a,\ x = -2$

Axial intercepts: (0, 0)

For C_2: $y = x + (a+2) + \dfrac{4}{x+2}$

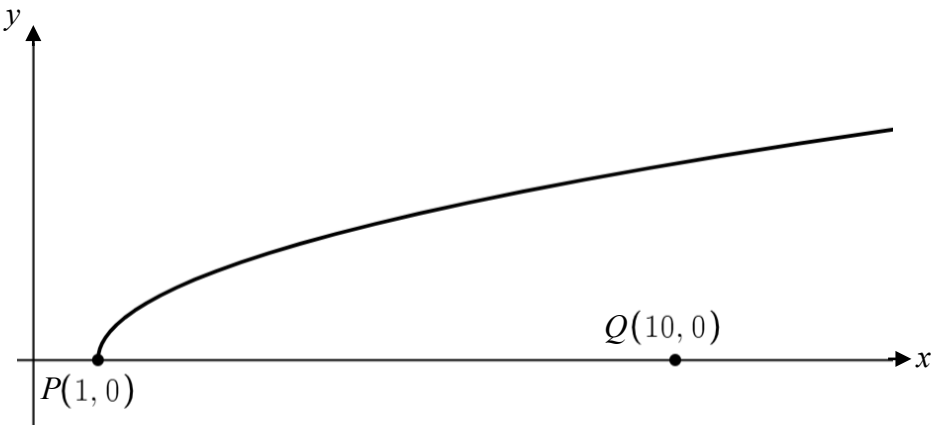
Equations of asymptotes: $y = x + (a+2),\ x = -2$

(d)

From the graph, for $\dfrac{ax}{x+2} > \dfrac{x^2 + (a+4)x + 2a + 8}{x+2}$

$x < -2$

10



A drone is programmed to fly from its launch pad at the point $P(1, 0)$ along a path described by the equation $y = \sqrt{x-1},\ x \geq 1$, as shown in the figure above. The pilot of the drone is located at the point $Q(10, 0)$.

(a) **(i)** Show that the distance, L, of the drone from the pilot as it travels along its flight path is given by $L = \sqrt{x^2 - 19x + 99}$. [1]

(ii) Use differentiation to find the exact minimum value of L, proving that it is a minimum. [4]

(b) The distance travelled by the drone along its flight path, denoted by F, is given by the integral

$$F(x) = \int_1^x \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x,$$

when the drone is at the point (x, y).

Given that the drone is flying at a constant speed of 0.8 units per second along its flight path, find the rate at which the shadow of the drone is moving horizontally when it is directly above its pilot. You may assume that the drone casts a shadow directly below it as it flies. [5]

Q10	Solution
(a)(i)	L = distance from $Q(10, 0)$ to a point $\left(x, \sqrt{x-1}\right)$ on the flight path. $L = \sqrt{(10-x)^2 + \left(\sqrt{x-1} - 0\right)^2} = \sqrt{x^2 - 19x + 99}$
(ii)	**Method 1** $\frac{dL}{dx} = \frac{1}{2}\left(x^2 - 19x + 99\right)^{-\frac{1}{2}}(2x - 99)$ Let $\frac{dL}{dx} = 0$ $\Rightarrow\ 2x - 19 = 0\ \Rightarrow\ x = \frac{19}{2}$ Using 1st derivative test

x	$\left(\frac{19}{2}\right)^-$	$\frac{19}{2}$	$\left(\frac{19}{2}\right)^+$
$\frac{dL}{dx}$	< 0	0	> 0
slope	↘	—	↗

Thus L is minimum when $x = \frac{19}{2}$

Hence the shortest distance from the drone to the pilot is

$= \sqrt{\left(\frac{19}{2}\right)^2 - 19\left(\frac{19}{2}\right) + 99} = \sqrt{\frac{35}{4}} = \frac{1}{2}\sqrt{35}$ units

Method 2

$L^2 = x^2 - 19x + 99$

$2L\frac{dL}{dx} = 2x - 19$ --- (1)

Let $\frac{dL}{dx} = 0$

$\Rightarrow\ 2x - 19 = 0\ \Rightarrow\ x = \frac{19}{2}$

Using 2nd derivative test

Differentiating (1): $2L\frac{d^2L}{dx^2} + 2\left(\frac{dL}{dx}\right)^2 = 2$

Substitute $\frac{dL}{dx} = 0$, $\frac{d^2L}{dx^2} = \frac{1}{L} > 0$ since $L > 0$

Thus L is minimum when $x = \frac{19}{2}$

Hence the shortest distance from the drone to the pilot is

	$= \sqrt{\left(\frac{19}{2}\right)^2 - 19\left(\frac{19}{2}\right) + 99} = \sqrt{\frac{35}{4}} = \frac{1}{2}\sqrt{35}$ units **Method 3** Since $L > 0$, when L is a minimum, L^2 will also be a minimum. $\frac{\mathrm{d}\left(L^2\right)}{\mathrm{d}x} = 2x - 19$ Let $= \frac{\mathrm{d}\left(L^2\right)}{\mathrm{d}x} = 0$ $\Rightarrow 2x - 19 = 0$ $\Rightarrow x = \frac{19}{2}$ Using 2nd derivative test, $\frac{\mathrm{d}^2\left(L^2\right)}{\mathrm{d}x} = 2 > 0$ Hence L^2 is a minimum $\Rightarrow L$ is a minimum. Thus L is minimum when $x = \frac{19}{2}$ Hence the shortest distance from the drone to the pilot is $= \sqrt{\left(\frac{19}{2}\right)^2 - 19\left(\frac{19}{2}\right) + 99} = \sqrt{\frac{35}{4}} = \frac{1}{2}\sqrt{35}$ units
(b)	Using chain rule, rate equation: $\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\mathrm{d}F}{\mathrm{d}x} \times \frac{\mathrm{d}x}{\mathrm{d}t}$ Given $F(x) = \int_1^x \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x$ By the reverse process of integration, $\frac{\mathrm{d}F}{\mathrm{d}x} = \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}$ Since $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2\sqrt{x-1}}$, we have $\frac{\mathrm{d}F}{\mathrm{d}x} = \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2} = \sqrt{1 + \frac{1}{4(x-1)}}$ For $\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\mathrm{d}F}{\mathrm{d}x} \times \frac{\mathrm{d}x}{\mathrm{d}t} = \sqrt{1 + \frac{1}{4(x-1)}} \cdot \frac{\mathrm{d}x}{\mathrm{d}t}$ Given the speed of drone, $\frac{\mathrm{d}F}{\mathrm{d}t} = 0.8$ and $x = 10$,

	$0.8 = \sqrt{1+\frac{1}{4(9)}} \cdot \frac{dx}{dt}$ $\Rightarrow \frac{dx}{dt} = \frac{24}{5\sqrt{37}}$ or 0.789 units /s

11 **(a)** A curve $y = f(x)$ passes through the point $\left(0, \frac{\pi}{3}\right)$, and has gradient given by

$$\frac{dy}{dx} = 1 + \cos y.$$

Using double angle formula, find an exact expression for y in terms of x. [4]

(b) An island is inhibited by two species of animals, A and B, and the populations of A and B are x and y respectively at time t. The rate of change of x and y are proportional to $x + y$ and $x - y$ respectively. The populations of A and B are initially the same at N. At the instant when $x = 2y$, the population of B is increasing at one-third the rate at which the population of A is increasing.

(i) Show that $\frac{dy}{dx} = \frac{x-y}{x+y}$. [2]

(ii) Using the substitution $y = vx$, show that the differential equation can be transformed to $x\frac{dv}{dx} = \frac{1-2v-v^2}{1+v}$. Hence solve the differential equation $\frac{dy}{dx} = \frac{x-y}{x+y}$. [8]

(iii) Find, in terms of N, the population of A at the instant when $x = 2y$. [1]

(iv) Explain whether the populations of A and B will be the same again? [2]

Q11	**Solution**
(a)	$\frac{dy}{dx} = 1 + \cos y$ $\int \frac{1}{1+\cos y}\,dy = \int 1\,dx$ $\int \frac{1}{1+\left(2\cos^2\frac{y}{2}-1\right)}\,dy = x + c$ $\frac{1}{2}\int \sec^2\frac{y}{2}\,dy = x + c$ $\frac{1}{2}.\frac{\tan\frac{y}{2}}{\left(\frac{1}{2}\right)} = x + d$ i.e. $\tan\frac{y}{2} = x + d$ When $x = 0, y = \frac{\pi}{3}$, $d = \tan\frac{\left(\frac{\pi}{3}\right)}{2} = \frac{1}{\sqrt{3}}$

	The solution is $\tan\frac{y}{2}=x+\frac{1}{\sqrt{3}}$ $y=2\tan^{-1}\left(x+\frac{1}{\sqrt{3}}\right)$
(b)(i)	$\frac{\mathrm{d}x}{\mathrm{d}t}=\alpha(x+y)$ where $\alpha\in\mathbb{R}$ $\frac{\mathrm{d}y}{\mathrm{d}t}=\beta(x-y)$ where $\beta\in\mathbb{R}$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)}{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)}=\frac{\beta(x-y)}{\alpha(x+y)}=k\left(\frac{x-y}{x+y}\right)$ where $k\in\mathbb{R}$ When $x=2y$, $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{3}$, $\frac{1}{3}=k\left(\frac{2y-y}{2y+y}\right)=\frac{1}{3}k \Rightarrow k=1$ Thus $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{x-y}{x+y}$ (shown)
(ii)	$y=vx$ --- (1) $\frac{\mathrm{d}y}{\mathrm{d}x}=v+x\frac{\mathrm{d}v}{\mathrm{d}x}$ --- (2) Subst (1) and (2): $v+x\frac{\mathrm{d}v}{\mathrm{d}x}=\frac{x-vx}{x+vx}$ $x\frac{\mathrm{d}v}{\mathrm{d}x}=\frac{1-v}{1+v}-v$ $=\frac{1-v-v(1+v)}{1+v}$ $=\frac{1-2v-v^2}{1+v}$ (shown) By variable separable,

	$\int \frac{1+v}{1-2v-v^2}\,\mathrm{d}v = \int \frac{1}{x}\,\mathrm{d}x$ $-\frac{1}{2}\int \frac{-2(1+v)}{1-2v-v^2}\,\mathrm{d}v = \ln x + c_1 \quad \text{since } x \geq 0$ $-\frac{1}{2}\ln\left\|1-2v-v^2\right\| = \ln x + c_2$ $\ln\left\|1-2v-v^2\right\| = -2\ln x - 2c_2$ $1-2v-v^2 = \pm e^{-2\ln x - 2c_2}$ $= \pm e^{-2c_2} e^{-2\ln x}$ $= Ax^{-2} \quad \text{where } A = \pm e^{-2c_2}$ $(1-2v-v^2)x^2 = A$ $1-2v-v^2 = \frac{A}{x^2}$ Substitute back $v = \frac{y}{x}$, $1-2\left(\frac{y}{x}\right)-\left(\frac{y}{x}\right)^2 = \frac{A}{x^2}$ $x^2 - 2xy - y^2 = A$ Initially, $x = y = N$, $N^2 - 2N^2 - N^2 = A \Rightarrow A = -2N^2$ The solution is $x^2 - 2xy - y^2 = -2N^2$
(iii)	When $x = 2y$, $x^2 - 2x\left(\frac{1}{2}x\right)-\left(\frac{1}{2}x\right)^2 = -2N^2$ $\frac{1}{4}x^2 = 2N^2$ $x = 2\sqrt{2}N \quad \text{since } x > 0$
(iv)	Let $x = y$ $x^2 - 2x^2 - x^2 = -2N^2$ $x^2 = N^2$ $x = N$ (initial state) Thus x will not equal y again. OR From part **(i)**, $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x-y}{x+y} < 1$ i.e. y increases at a slower rate than x. As the initial populations are equal, x will not equal y again.